

EDUCATION

Massachusetts Institute of Technology - Cambridge, MA

September 2023 - Expected May 2027

Bachelor of Science in Mathematics and Computer Science

Cumulative GPA: 4.8/5

Relevant Courses: Undergraduate Math (Algebra, Analysis, Topology, Machine Learning (Deep Learning, Statistical ML), Differential Geometry, Statistics), Statistics (Inference, Information Theory, non-Asymptotic stats), Computer Science (Programming, Algorithms, Theoretical Computer Science), Graduate Mathematics (Commutative Algebra, Algebraic Topology/Geometry, Number Theory),

AWARDS

IMO (International Mathematical Olympiad) Gold Medalist

July 2022

EXPERIENCE

AI Safety Research

June 2026 – August 2026

Visiting Researcher, Alignment Research Center

Berkeley, CA

- Researched heuristic estimators for predicting the output of random multi-layer perceptrons (MLPs)
- Built and empirically tested estimation algorithms against baseline methods
- Investigated AI safety and the trustworthiness of modern machine learning systems

Circuit Complexity Research

September 2025 – Present

Researcher at MIT under Daniel Reichmann

Cambridge, MA

- Studied circuit complexity, including lower and upper bounds for computing Boolean functions
- Designed circuit constructions implementing various functions to probe the fundamental limits of computation

Statistical ML Research

June 2025 – August 2025

SPUR Program under Ankur Moitra

Cambridge, MA

- Research in Statistical Machine Learning on Graph retrieval
- Obtained new algorithm with improved sample complexity on Gaussian Graphical models
- Accepted to ICML 2026

Machine Learning Algorithm Design

August 2025 – September 2025

Intern at Baichuan AI

China

- Designed and implemented Loss functions for training Transformer models
- Experimented with model structure for Time series prediction on medical data
- Improved upon current loss implementation to account for time and value discrepancy

Global Teaching Lab

January 2024

Instructor

Kigali, Rwanda

- Instructed 40 local teachers and few hundred students to prepare for International Mathematics Olympiad
- Promoted Rwandan Mathematics Program to six local high schools and increased student involvement in mathematics competitions
- Inspired students to pursue mathematics beyond the traditional classroom setting

FABRIC

July 2024 – August 2024

PAIR (Program on AI and Reasoning) participant

Somerset, UK

- Learnt about modern neural network structures such as CNN, Transformers, Diffusion models and more
- Experimented with mechanistic interpretability and steering vectors
- Experimented with safety against prompt injection

Mathink

May 2023 – Present

Main Course Designer and Teacher

NSW, Australia

- Designed curriculum to bridge students to higher level mathematics, usually University content
- Remotely taught and designed courses for over 3 years, knows how to teach well

SKILLS & INTERESTS

(Programming)Skills: Some

Interests: Food, outdoors sometimes, a bunch of other stuff if you ask me about it